Andreev transport through a ferromagnet–bilayer-graphene–superconductor junction

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We study Andreev transport through a hybrid graphene junction composed of a ferromagnetic monolayer lead, a finite normal AB-stacked bilayer-graphene region, and an s-wave superconducting monolayer lead. Two atomically distinct contact geometries are considered, depending on whether the superconducting lead continues from the same graphene layer as the ferromagnetic lead or from the opposite layer. Starting from a microscopic tight-binding Hamiltonian, we formulate a Bogoliubov–de Gennes transfer-matrix approach in which the finite bilayer is propagated through flux-normalized modes and the two contacts are treated by local atomic matching conditions. The resulting conductance can be written in terms of a four-dimensional round-trip matrix whose eigenvalues diagnose the finite-length resonance structure responsible for the bright ridges observed in conductance maps. This manuscript draft emphasizes the compact model and formalism; numerical results and ridge overlays will be inserted in the next stage.

I. INTRODUCTION

Andreev reflection converts an incident electron into a reflected hole while transferring a Cooper pair to a superconducting condensate. In graphene-based systems this process is sensitive not only to the superconducting gap, but also to the band structure, transverse momentum, doping, and interface geometry. Adding a ferromagnetic region introduces spin-dependent longitudinal momenta and therefore suppresses or reshapes Andreev reflection through the mismatch between the electron and the time-reversed hole channel.

The system considered here combines these ingredients with a finite AB-stacked bilayer-graphene section. The bilayer acts as an internal coherent cavity: an electron-like or hole-like excitation injected from the ferromagnetic lead can undergo repeated normal and Andreev reflections at the two atomic contacts before escaping. The finite length N then enters through propagation phases and evanescent factors, producing resonance ridges in maps of the conductance as a function of excitation energy and bilayer length.

The aim of this paper is to formulate a microscopic transfer-matrix description of this problem and to use it to interpret the length-dependent Andreev structures. Unlike a phenomenological BTK model with an adjustable barrier strength [?], the contact transparency is determined here by the actual honeycomb connectivity, the layer termination, the exchange field, the chemical potentials, and the superconducting pairing. The normal bilayer transfer construction follows the lattice convention of Ref. ? , generalized here to a spin-resolved Bogoliubov–de Gennes problem.

The central formal result is a round-trip representation of the scattering problem. For each transverse channel K, spin sector σ, and geometry g, repeated internal cycles are summed by the resolvent

$$\left[\mathbb{I}_4 - \mathsf{M}_{\mathrm{rt}}^{(g)}(E, K, \sigma, N)\right]^{-1}, \tag{1}$$

where $\mathsf{M}_{\mathrm{rt}}^{(g)}$ is the microscopic round-trip matrix of the finite bilayer cavity. Its eigenvalues provide a direct diagnostic of the bright ridges in the conductance maps: a fixed-K resonance occurs when a bright round-trip eigenvalue approaches unity, while a ridge in the transverse-integrated conductance survives only when the corresponding resonance branch is sufficiently flat over the open incident-channel window.

The manuscript is organized as follows. Section **??** defines the lattice geometry, Hamiltonian, and BdG sector convention. Section **??** constructs the transfer matrices, flux normalization, and stable finite-length propagation. Section **??** derives the interface maps, round-trip matrix, scattering amplitudes, and conductance. Section **??** explains the resonance diagnostics used to interpret the conductance ridges. Section **??** is reserved for the numerical maps and overlays.

II. MODEL

A. Geometry and physical domains

The system is illustrated in Fig. **??**. It is composed of two graphene layers, labeled by $j = 1, 2$, arranged in two contact geometries denoted by $1 \to 1$ and $1 \to 2$, corresponding to Figs. **??**(a) and **??**(b), respectively. The longitudinal and transverse directions are indexed by the integers m and n. With this convention, the finite bilayer region occupies

$$1 \leq m \leq N \tag{2}$$

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in both geometries. For $m \le 0$ the graphene monolayer is ferromagnetic, whereas for $m \ge N+1$ the monolayer is an s-wave superconductor.

The main difference between the two geometries concerns the right contact, where the finite bilayer is coupled to the superconducting monolayer. In the $1\to 1$ geometry, the superconductor is coupled to the lower layer, the same layer that is connected to the ferromagnetic monolayer. In the $1\to 2$ geometry, the ferromagnetic monolayer is still connected to the lower layer, but the superconducting monolayer is coupled to the upper layer. This distinction is most clearly visualized in the block diagram of Fig. **??**(d). In both geometries we assume Bernal stacking in the central region, as shown in Fig. **??**(c), with an A atom of the lower layer coupled to a B atom of the upper layer by the interlayer hopping $t_\perp$.

We now translate this figure-based description into the domains used in the Hamiltonian sums. The set of layer–cell pairs that physically exist in the $1\to 1$ geometry is

$$\mathcal{D}_{1\to 1} = \{(1,m) : m \in \mathbb{Z}\} \cup \{(2,m) : 1 \le m \le N\}. \tag{3}$$

Thus layer 1 extends through the whole junction, while layer 2 exists only in the finite bilayer overlap. For the $1\to 2$ geometry,

$$\mathcal{D}_{1\to 2} = \{(1,m) : m \le N\} \cup \{(2,m) : m \ge 1\}. \tag{4}$$

Thus layer 1 extends from the ferromagnetic lead through the central bilayer and terminates at the right edge, while layer 2 begins at the left edge of the central bilayer and continues into the superconducting lead. In what follows, $g \in \{1\to 1, 1\to 2\}$ labels the chosen geometry and $\mathcal{D}_g$ denotes the corresponding physical domain.

All microscopic sums below are restricted to $(j,m) \in \mathcal{D}_g$. A hopping or pairing term is included only if every site appearing in that term belongs to the physical domain. A missing neighbor at a terminated layer is therefore not assigned a zero amplitude by hand; the corresponding bond is absent from the Hamiltonian. This convention is the microscopic origin of the termination conditions used later in the interface matching.

The remaining internal labels are the sublattice index $\alpha = A, B$ and the spin label $\sigma = \uparrow, \downarrow$, with

$$s_\uparrow = +1, \qquad s_\downarrow = -1. \tag{5}$$

B. Real-space tight-binding Hamiltonian

The microscopic Hamiltonian is written as

$$\hat{H}^{(g)} = \hat{H}_t^{(g)} + \hat{H}_\perp + \hat{H}_{\text{on}}^{(g)} + \hat{H}_\Delta^{(g)}. \tag{6}$$

The superscript (g) emphasizes that the intralayer, onsite, and pairing terms depend on the physical domain of the chosen contact geometry.

With the honeycomb convention used in Ref. **?** , each A site is connected to three nearest-neighbor B sites. The intralayer hopping is

$$\hat{H}_t^{(g)} = -t \sum_{(j,m)\in\mathcal{D}_g} \sum_{n,\sigma} \Big[\hat{a}^\dagger_{jmn\sigma}\hat{b}_{jmn\sigma} + \hat{a}^\dagger_{jmn\sigma}\hat{b}_{j,m+1,n,\sigma} + \hat{a}^\dagger_{jmn\sigma}\hat{b}_{j,m,n+1,\sigma} + \text{H.c.}\Big]. \tag{7}$$

The longitudinal bond in Eq. (**??**) is present only when both (j,m) and $(j,m+1)$ belong to $\mathcal{D}_g$.

The AB interlayer hopping exists only inside the central bilayer, where both layers are present:

$$\hat{H}_\perp = -t_\perp \sum_{m=1}^{N} \sum_{n,\sigma} \Big(\hat{a}^\dagger_{1mn\sigma}\hat{b}_{2mn\sigma} + \text{H.c.}\Big). \tag{8}$$

Thus the dimer bond connects A_1 to B_2 for $1 \le m \le N$.

The onsite normal term is

$$\hat{H}_{\text{on}}^{(g)} = \sum_{(j,m)\in\mathcal{D}_g} \sum_{n,\alpha,\sigma} \left[V_j(m) - \mu(m) - s_\sigma h(m)\right] \hat{n}_{jmn\alpha\sigma}, \tag{9}$$

where

$$\hat{n}_{jmn\alpha\sigma} = \hat{c}^\dagger_{jmn\alpha\sigma}\hat{c}_{jmn\alpha\sigma}. \tag{10}$$

Here $\alpha = A, B$ is the sublattice index. The spatial region is determined by m and by the physical domain $\mathcal{D}_g$, not by an additional operator label. The chemical potential is piecewise constant,

$$\mu(m) = \begin{cases} \mu_F, & m \le 0, \\ \mu_C, & 1 \le m \le N, \\ \mu_S, & m \ge N+1, \end{cases} \tag{11}$$

and the collinear exchange field is confined to the ferromagnetic lead,

$$h(m) = \begin{cases} h, & m \le 0, \\ 0, & m \ge 1. \end{cases} \tag{12}$$

The layer potentials $V_1(m)$ and $V_2(m)$ are nonzero only in the central bilayer and allow for layer-resolved gating.

The superconducting lead contains onsite spin-singlet pairing,

$$\hat{H}_\Delta^{(g)} = \sum_{(j,m)\in\mathcal{D}_g} \sum_{n,\alpha} \Big[\Delta(m)\hat{c}^\dagger_{jmn\alpha\uparrow}\hat{c}^\dagger_{jmn\alpha\downarrow} + \Delta^*(m)\hat{c}_{jmn\alpha\downarrow}\hat{c}_{jmn\alpha\uparrow}\Big]. \tag{13}$$

where

$$\Delta(m) = \begin{cases} 0, & m \le N, \\ \Delta_0 e^{i\varphi}, & m \ge N+1. \end{cases} \tag{14}$$

Because the sum in Eq. (**??**) is restricted to $\mathcal{D}_g$, the pair potential is placed only on the layer that physically continues into the superconducting lead.

C. Fourier-space Hamiltonian

We now use translational invariance along the transverse direction to write the model in momentum space. For N_y transverse cells with periodic boundary conditions, define the dimensionless transverse momentum $K = k_y a_y \in [-\pi, \pi]$ and use

$$\hat{a}_{jmn\sigma} = \frac{1}{\sqrt{N_y}} \sum_K e^{iKn} \hat{a}_{jmK\sigma}, \qquad \hat{b}_{jmn\sigma} = \frac{1}{\sqrt{N_y}} \sum_K e^{iKn} \hat{b}_{jmK\sigma}. \tag{15}$$

The two intralayer bonds that do not change m combine into the effective intracell hopping

$$\eta(K) = t(1 + e^{iK}), \qquad \xi = t, \tag{16}$$

where ξ denotes the longitudinal hopping across one transfer step. The intralayer term becomes

$$\hat{H}_t^{(g)}(K) = -\sum_{(j,m)\in\mathcal{D}_g} \sum_\sigma \Big[\eta(K) \hat{a}^\dagger_{jmK\sigma} \hat{b}_{jmK\sigma} + \xi \hat{a}^\dagger_{jmK\sigma} \hat{b}_{j,m+1,K,\sigma} + \text{H.c.} \Big]. \tag{17}$$

As in real space, the longitudinal bond is included only when the two sites connected by it exist in $\mathcal{D}_g$.

The interlayer hopping is diagonal in K because the dimer bond is vertical within the same cell:

$$\hat{H}_\perp(K) = -t_\perp \sum_{m=1}^{N} \sum_\sigma \left(\hat{a}^\dagger_{1mK\sigma} \hat{b}_{2mK\sigma} + \text{H.c.} \right). \tag{18}$$

The onsite normal term is also diagonal in K,

$$\hat{H}_{\text{on}}^{(g)}(K) = \sum_{(j,m)\in\mathcal{D}_g} \sum_{\alpha,\sigma} [V_j(m) - \mu(m) - s_\sigma h(m)] \times \hat{n}_{jmK\alpha\sigma}. \tag{19}$$

The pairing term is different: because it contains two creation or two annihilation operators at the same transverse cell, it couples opposite transverse momenta,

$$\hat{H}_\Delta^{(g)}(K) = \sum_{(j,m)\in\mathcal{D}_g} \sum_\alpha \Big[\Delta(m) \hat{c}^\dagger_{jmK\alpha\uparrow} \hat{c}^\dagger_{jm,-K,\alpha\downarrow} + \Delta^*(m) \hat{c}_{jm,-K,\alpha\downarrow} \hat{c}_{jmK\alpha\uparrow} \Big]. \tag{20}$$

Thus the normal part is diagonal in K, while the singlet-pairing part closes only after K and $-K$ are treated together. The independent scattering channel labeled by K is therefore the BdG sector

$$(K, \sigma) \oplus (-K, -\sigma), \tag{21}$$

not an ordinary normal-state single-momentum block. In this sense the Hamiltonian is block diagonal in paired BdG sectors; no additional independent sum over $-K$ is introduced when the scattering problem is solved at fixed K.

III. BOGOLIUBOV–DE GENNES EQUATIONS

The Fourier-space Hamiltonian of Sec. **??** is now converted into algebraic tight-binding equations. This intermediate step is useful because it shows explicitly how the microscopic model becomes the transfer recurrence used in Sec. **??**.

A. BdG sector and quasiparticle operator

For a fixed electron spin σ, the corresponding hole has spin $-\sigma$. We use the quasiparticle operator

$$\hat{\gamma}^\dagger_{EK\sigma} = \sum_{j,m,\alpha} \Big[u_{j\alpha}(m) \hat{c}^\dagger_{jmK\alpha\sigma} + s_\sigma v_{j\alpha}(m) \hat{c}_{jm,-K,\alpha,-\sigma} \Big]. \tag{22}$$

The factor s_σ in the hole component cancels the spin-dependent sign produced by the spin-singlet pairing commutator. With this convention, the two independent sectors $(e_\uparrow, h_\downarrow)$ and $(e_\downarrow, h_\uparrow)$ have the same off-diagonal BdG entries $+\Delta$ and $+\Delta^*$.

The eigenoperator equation

$$[\hat{H}^{(g)}, \hat{\gamma}^\dagger_{EK\sigma}] = E \hat{\gamma}^\dagger_{EK\sigma} \tag{23}$$

gives, in compact one-particle notation,

$$\begin{pmatrix} H_\sigma(K) & \Delta \\ \Delta^\dagger & -H^*_{-\sigma}(-K) \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = E \begin{pmatrix} u \\ v \end{pmatrix}. \tag{24}$$

The lower block contains three operations: spin reversal, momentum reversal, and complex conjugation. These operations are responsible for the signs in the hole equations below.

B. Monolayer BdG equations

For a uniform monolayer region with parameters (μ, h, Δ) and no layer gate potential, define

$$\epsilon_{e,\sigma} = E + \mu + s_\sigma h, \qquad \epsilon_{h,\sigma} = E - \mu + s_\sigma h. \tag{25}$$

Projecting Eq. (**??**) onto the A and B sublattices gives

$$\epsilon_{e,\sigma} u_A(m) = -\eta u_B(m) - \xi u_B(m+1) + \Delta v_A(m), \tag{26}$$

$$\epsilon_{e,\sigma} u_B(m) = -\eta^* u_A(m) - \xi^* u_A(m-1) + \Delta v_B(m), \tag{27}$$

$$\epsilon_{h,\sigma} v_A(m) = +\eta v_B(m) + \xi v_B(m+1) + \Delta^* u_A(m), \tag{28}$$

$$\epsilon_{h,\sigma} v_B(m) = +\eta^* v_A(m) + \xi^* v_A(m-1) + \Delta^* u_B(m). \tag{29}$$

The electron equations inherit the minus sign of the hopping Hamiltonian, whereas the hole equations have the opposite hopping sign because they come from the block $-H^*_{-\sigma}(-K)$. The exchange field enters both $\epsilon_{e,\sigma}$ and $\epsilon_{h,\sigma}$ with the same sign because the hole is a missing electron of spin $-\sigma$.

In the ferromagnetic lead $\Delta = 0$ and $h \neq 0$. In the superconducting lead $h = 0$ and $\Delta = \Delta_0 e^{i\varphi}$.

C. Central bilayer equations

Inside the central bilayer, $h = \Delta = 0$. Define the layer-dependent scalar parameters

$$x_{e,j} = E + \mu_C - V_j, \qquad x_{h,j} = \mu_C - E - V_j. \tag{30}$$

The electron amplitudes satisfy

$$x_{e,1}u_{A1}(m) = -\eta u_{B1}(m) - \xi u_{B1}(m+1) - t_\perp u_{B2}(m), \tag{31}$$

$$x_{e,1}u_{B1}(m) = -\eta^* u_{A1}(m) - \xi^* u_{A1}(m-1), \tag{32}$$

$$x_{e,2}u_{A2}(m) = -\eta u_{B2}(m) - \xi u_{B2}(m+1), \tag{33}$$

$$x_{e,2}u_{B2}(m) = -\eta^* u_{A2}(m) - \xi^* u_{A2}(m-1) - t_\perp u_{A1}(m). \tag{34}$$

The interlayer hopping appears only in the A_1 and B_2 equations, reflecting the AB dimer bond shown in Fig. **??**(c).

The central hole equations have the same algebraic structure after the replacement

$$u \mapsto v, \qquad x_{e,j} \mapsto x_{h,j}. \tag{35}$$

This is the reason the central BdG transfer matrix in the next section is a direct sum of two normal bilayer transfer matrices evaluated at the two scalar arguments $\mu_C + E$ and $\mu_C - E$.

IV. TRANSFER-MATRIX AND MODE FORMALISM

The lattice equations of Sec. **??** are second-order recurrences in the longitudinal index: an equation at cell m contains amplitudes at $m-1$, m, and $m+1$. The transfer-matrix construction rewrites them as first-order recurrences by grouping amplitudes on the two sides of a longitudinal cut.

A. Staggered transfer states

For a normal monolayer, the natural transfer state is not $(A(m), B(m))^T$ but the staggered pair

$$\phi_m = \begin{pmatrix} A(m-1) \\ B(m) \end{pmatrix}, \qquad \phi_{m+1} = \begin{pmatrix} A(m) \\ B(m+1) \end{pmatrix}. \tag{36}$$

This choice is dictated by Eqs. (**??**) and (**??**): the A equation contains $B(m+1)$, while the B equation contains $A(m-1)$.

For the normal bilayer, the corresponding state is

$$\Psi_m = \begin{pmatrix} A_1(m-1) \\ B_1(m) \\ A_2(m-1) \\ B_2(m) \end{pmatrix}, \qquad \Psi_{m+1} = \begin{pmatrix} A_1(m) \\ B_1(m+1) \\ A_2(m) \\ B_2(m+1) \end{pmatrix}. \tag{37}$$

The central BdG transfer state is the direct electron–hole extension

$$\Xi_m = \begin{pmatrix} u_{A1}(m-1) \\ u_{B1}(m) \\ u_{A2}(m-1) \\ u_{B2}(m) \\ v_{A1}(m-1) \\ v_{B1}(m) \\ v_{A2}(m-1) \\ v_{B2}(m) \end{pmatrix}. \tag{38}$$

The same staggered ordering is used everywhere below; changing it would also change the signs and the selector components used at the interfaces.

B. From tight-binding equations to transfer matrices

The connection between the algebraic tight-binding equations and the transfer matrix can be seen directly in the central bilayer. Let $x_j = \epsilon - V_j$, where ϵ is the scalar energy argument of the normal bilayer problem. Rearranging Eqs. (**??**)–(**??**) with the amplitudes belonging to Ψ_{m+1} on the left and those belonging to Ψ_m on the right gives

$$L_{\mathrm{BL}}\Psi_{m+1} = R_{\mathrm{BL}}\Psi_m, \qquad T_{\mathrm{BL}}(\epsilon, K) = L_{\mathrm{BL}}^{-1}R_{\mathrm{BL}}. \tag{39}$$

The matrices are

$$L_{\mathrm{BL}} = \begin{pmatrix} x_1 & \xi & 0 & 0 \\ \eta^* & 0 & 0 & 0 \\ 0 & 0 & x_2 & \xi \\ t_\perp & 0 & \eta^* & 0 \end{pmatrix}, \tag{40}$$

$$R_{\mathrm{BL}} = \begin{pmatrix} 0 & -\eta & 0 & -t_\perp \\ -\xi^* & -x_1 & 0 & 0 \\ 0 & 0 & 0 & -\eta \\ 0 & 0 & -\xi^* & -x_2 \end{pmatrix}. \tag{41}$$

The first row of Eq. (**??**), for example, is just Eq. (**??**) written as

$$x_1 A_1(m) + \xi B_1(m+1) = -\eta B_1(m) - t_\perp B_2(m). \tag{42}$$

Thus the transfer matrix is not an additional approximation; it is a reorganization of the same tight-binding equations.

The normal bilayer modes may be written in Bloch form. The corresponding transfer roots are

$$\lambda_{\nu,\pm} = \exp\left[i\left(\frac{K}{2} \pm Q_\nu\right)\right], \tag{43}$$

where Q_ν is determined by the bilayer secular equation. In the numerical implementation the eigenvectors are obtained from the transfer matrix itself rather than from closed-form spinors, because special points such as layer charge neutrality may make intermediate analytic divisions singular.

C. Central BdG transfer matrix

Since the central region is normal, electron and hole amplitudes propagate independently. Using Eq. (**??**), the central BdG transfer matrix is

$$\Xi_{m+1} = T_C(E,K)\Xi_m, \tag{44}$$
$$T_C(E,K) = \mathrm{diag}(T_e, T_h), \tag{45}$$
$$T_e = T_{\mathrm{BL}}(\mu_C + E, K), \tag{46}$$
$$T_h = T_{\mathrm{BL}}(\mu_C - E, K).$$

The explicit determinant checks and limiting cases of T_{BL} are collected in Appendix **??**. Lead transfer matrices T_F and T_S are constructed from the monolayer BdG equations, Eqs. (**??**)–(**??**), by the same staggered procedure.

D. Flux metric and mode classification

The propagation direction of a multiband transfer eigenmode must be determined by its lattice current, not by the sign of a phase label alone. For the monolayer BdG ordering

$$\Phi_m = \big(u_A(m-1) \;\; u_B(m) \;\; v_A(m-1) \;\; v_B(m)\big)^T, \tag{47}$$

the quasiparticle current can be written as the bilinear form

$$J_{\mathrm{qp}}(\Phi) = \Phi^\dagger \Sigma \Phi, \tag{48}$$
$$\Sigma = \begin{pmatrix} \sigma_\xi & 0 \\ 0 & -\sigma_\xi \end{pmatrix}, \qquad \sigma_\xi = \begin{pmatrix} 0 & -i\xi \\ i\xi^* & 0 \end{pmatrix}.$$

For the central ordering of Eq. (**??**),

$$\Sigma_C = \mathrm{diag}(\sigma_\xi, \sigma_\xi, -\sigma_\xi, -\sigma_\xi). \tag{49}$$

Current conservation in a uniform region implies the pseudo-unitarity relation

$$T^\dagger \Sigma T = \Sigma. \tag{50}$$

If $T\zeta_i = \lambda_i \zeta_i$ and $T\zeta_j = \lambda_j \zeta_j$, Eq. (**??**) gives

$$(\lambda_i^* \lambda_j - 1)\, \zeta_i^\dagger \Sigma \zeta_j = 0. \tag{51}$$

Thus distinct propagating modes can be chosen flux-orthogonal. Degenerate propagating subspaces should be explicitly orthogonalized with respect to Σ before squared modal amplitudes are interpreted as probabilities.

In the left ferromagnetic lead, the prescribed incident electron is a unit-flux mode ζ_{in} with $J > 0$. The outgoing lead basis Z_F^- contains reflected propagating modes with $J < 0$ and evanescent modes decaying toward $m \to -\infty$. At the left boundary,

$$\Phi_F(1) = \zeta_{\mathrm{in}} + Z_F^- r. \tag{52}$$

In the right superconducting lead, the physical basis Z_S^+ contains propagating modes with $J > 0$ and evanescent modes decaying toward $m \to +\infty$:

$$\Phi_S(N+1) = Z_S^+ t. \tag{53}$$

E. Stable finite-length propagation

The finite central bilayer must retain all eight transfer modes. Let V_+ and V_- denote the 8×4 matrices whose columns are the right-going/right-decaying and left-going/left-decaying central modes. The central boundary states are represented as

$$\Xi_L = V_+ a_+^L + V_- a_-^L, \tag{54}$$
$$\Xi_R = V_+ P_+(N) a_+^L + V_- a_-^R. \tag{55}$$

Here

$$P_+(N) = \mathrm{diag}\left[(\lambda_1^+)^N, \ldots, (\lambda_4^+)^N\right], \tag{56}$$
$$P_-(N) = \mathrm{diag}\left[(\lambda_1^-)^{-N}, \ldots, (\lambda_4^-)^{-N}\right]. \tag{57}$$

The modes are referenced at the boundary where their explicit propagation factor has modulus no greater than one. This representation is analytically equivalent to direct propagation through T_C^N, but avoids generating exponentially large factors from evanescent modes.

V. INTERFACE MAPS AND CONDUCTANCE

A. Local interface maps

At each atomically clean contact, the layer that continues into the monolayer lead is matched by continuity, while the layer that terminates imposes the absence of the fictitious neighbor outside the physical domain. In the modal basis this produces one local 6×6 linear system at each interface. Solving the left interface gives

$$a_+^L = t_{\mathrm{in}} + R_F a_-^L, \tag{58}$$

and

$$r = r^{(0)} + T_{F \leftarrow C} a_-^L. \tag{59}$$

Here t_{in} is the central amplitude injected by the incident ferromagnetic electron, R_F reflects a returning central mode back into the bilayer, and $T_{F\leftarrow C}$ extracts a central mode into outgoing ferromagnetic lead amplitudes.

At the superconducting contact,

$$a_-^R = R_S^{(g)} a_+^R, \qquad t = T_{S\leftarrow C}^{(g)} a_+^R. \tag{60}$$

The geometry index g enters only through the right-contact selectors, because the right lead continues from layer 1 in the $1 \to 1$ geometry and from layer 2 in the $1 \to 2$ geometry. The matrix $R_S^{(g)}$ contains both normal and Andreev blocks in central modal space,

$$R_S^{(g)} = \begin{pmatrix} R_S^{ee,(g)} & R_S^{eh,(g)} \\ R_S^{he,(g)} & R_S^{hh,(g)} \end{pmatrix}. \tag{61}$$

The off-diagonal blocks arise from the Nambu structure of the superconducting lead modes; no phenomenological Andreev phase is inserted by hand.

B. Round-trip matrix

A right-going central amplitude leaving the left boundary first propagates to the right boundary, reflects at the superconducting contact, propagates back, and reflects at the ferromagnetic contact. The corresponding round-trip matrix is

$$\mathsf{M}_{\mathrm{rt}}^{(g)}(E,K,\sigma,N) = R_F P_-(N) R_S^{(g)} P_+(N). \tag{62}$$

The exact cavity equation is

$$\left[\mathbb{I}_4 - \mathsf{M}_{\mathrm{rt}}^{(g)}(E,K,\sigma,N)\right] a_+^L = t_{\mathrm{in}}. \tag{63}$$

All matrices in Eq. (**??**) are 4×4, while t_{in} and a_+^L are four-component central modal vectors.

Solving Eq. (**??**) and inserting the result into Eq. (**??**) gives the Andreev-reflected amplitude

$$\begin{aligned} r_{he}^{(g)} &= e_h^\dagger T_{F\leftarrow C} P_-(N) R_S^{(g)} P_+(N) \\ &\times \left[\mathbb{I}_4 - \mathsf{M}_{\mathrm{rt}}^{(g)}\right]^{-1} t_{\mathrm{in}}. \end{aligned} \tag{64}$$

Here $e_h = (0,1)^T$ selects the hole component of the outgoing ferromagnetic lead amplitude. The direct reflection term at the left contact has no hole component because the left lead is normal.

The quasiparticle transmission into the superconducting lead is

$$t^{(g)} = T_{S\leftarrow C}^{(g)} P_+(N) \left[\mathbb{I}_4 - \mathsf{M}_{\mathrm{rt}}^{(g)}\right]^{-1} t_{\mathrm{in}}. \tag{65}$$

For $|E| < \Delta_0$, the physical superconducting modes are evanescent and carry no asymptotic transmitted flux.

C. BTK conductance

With all propagating lead modes normalized to unit flux, the reflection and transmission probabilities are

$$R_N = |r_{ee}|^2, \qquad R_A = |r_{he}|^2, \qquad T_{\mathrm{QP}} = \sum_{\nu\in\mathrm{prop.}} |t_\nu|^2. \tag{66}$$

Flux conservation gives

$$R_N + R_A + T_{\mathrm{QP}} = 1 \tag{67}$$

on the support of the incident channel. The single-channel BTK conductance is

$$\begin{aligned} g(E,K,\sigma,N) = \chi_{\mathrm{in}}(E,K,\sigma)\Big[1 &- R_N(E,K,\sigma,N) \\ &+ R_A(E,K,\sigma,N)\Big]. \end{aligned} \tag{68}$$

where

$$\chi_{\mathrm{in}}(E,K,\sigma) = \begin{cases} 1, & \text{incoming electron channel open,} \\ 0, & \text{otherwise.} \end{cases} \tag{69}$$

The condition $\chi_{\mathrm{in}} = 1$ means that a propagating electron eigenmode of T_F with $J>0$ exists at the chosen (E,K,σ). Equivalently, using Eq. (**??**),

$$\begin{aligned} g(E,K,\sigma,N) = \chi_{\mathrm{in}}(E,K,\sigma)\Big[&2R_A(E,K,\sigma,N) \\ &+ T_{\mathrm{QP}}(E,K,\sigma,N)\Big]. \end{aligned} \tag{70}$$

The dimensionless conductance per transverse unit cell, in units of e^2/h, is. In Eq. (**??**), $R_A^{(g)}$ and $T_{\mathrm{QP}}^{(g)}$ are evaluated at the same arguments (E,K,σ,N) as χ_{in}.

$$\begin{aligned} G^{(g)}(E,N) = \frac{1}{2\pi}\sum_{\sigma=\pm1}&\int_{-\pi}^{\pi} dK\, \chi_{\mathrm{in}}(E,K,\sigma) \\ &\times \left[2R_A^{(g)} + T_{\mathrm{QP}}^{(g)}\right]. \end{aligned} \tag{71}$$

For subgap energies,

$$|E| < \Delta_0, \qquad T_{\mathrm{QP}}^{(g)} = 0, \tag{72}$$

and hence

$$\begin{aligned} G_A^{(g)}(E,N) = \frac{1}{2\pi}\sum_{\sigma=\pm1}&\int_{-\pi}^{\pi} dK\, \chi_{\mathrm{in}}(E,K,\sigma) \\ &\times 2\left|r_{he}^{(g)}(E,K,\sigma,N)\right|^2. \end{aligned} \tag{73}$$

For a system of N_y transverse cells, the total conductance is $(e^2/h)N_y G^{(g)}$, and the conductance per physical width $W = N_y a_y$ is $(e^2/h)G^{(g)}/a_y$.

VI. RESONANCE INTERPRETATION

A. Real-axis resonance diagnostics

The exact object controlling internal resonances is the round-trip matrix in Eq. (**??**). Since the scattering amplitudes contain the resolvent $[\mathbb{I}_4 - \mathsf{M}_{\mathrm{rt}}^{(g)}]^{-1}$, a resonance is associated with near-singularity of

$$\mathcal{R}^{(g)}(E,K,\sigma,N) = \mathbb{I}_4 - \mathsf{M}_{\mathrm{rt}}^{(g)}(E,K,\sigma,N). \tag{74}$$

The determinant

$$D_g(E,K,\sigma,N) = \det \mathcal{R}^{(g)}(E,K,\sigma,N) \tag{75}$$

is useful analytically, but on the real energy axis the more robust numerical diagnostic is the smallest singular value

$$s_{\min}^{(g)}(E,K,\sigma,N) = s_{\min}\left[\mathcal{R}^{(g)}(E,K,\sigma,N)\right]. \tag{76}$$

A small value of $s_{\min}$ indicates that a pole of the open scattering problem lies close to the real axis. It does not by itself guarantee a visible peak in the conductance, because the corresponding pole may be weakly excited or weakly read out as a hole.

B. Round-trip eigenvalues and bright versus dark poles

Assume for the moment that $\mathsf{M}_{\rm rt}$ is diagonalizable. Let

$$\begin{aligned} \mathsf{M}_{\rm rt} v_j^R &= \mu_j v_j^R, \\ (v_j^L)^\dagger \mathsf{M}_{\rm rt} &= \mu_j (v_j^L)^\dagger, \\ (v_j^L)^\dagger v_k^R &= \delta_{jk}. \end{aligned} \tag{77}$$

Then the resolvent gives the modal decomposition

$$r_{he}^{(g)} = \sum_j \frac{C_j^{(g)}}{1-\mu_j^{(g)}}. \tag{78}$$

The residue factor is

$$C_j^{(g)} = \left[e_h^\dagger T_{F\leftarrow C} P_- R_S^{(g)} P_+ v_j^R\right]\left[(v_j^L)^\dagger t_{\rm in}\right]. \tag{79}$$

Writing

$$\mu_j = \rho_j e^{i\Theta_j}, \tag{80}$$

ρ_j measures the survival of the amplitude after one complete internal cycle, while Θ_j is the coherent round-trip phase. The factor C_j is the product of an injection weight and a hole-readout weight. Thus a pole-like feature of $\mathsf{M}_{\rm rt}$ is bright in $G(E,N)$ only if $|C_j|$ is appreciable. A mode for which $\mu_j \simeq 1$ but $C_j \simeq 0$ is a dark pole: it is present in the internal cavity but nearly invisible in the measured Andreev conductance.

For a single isolated bright mode, the approximate fixed-channel resonance condition is

$$\begin{aligned} \rho_j(E,K,\sigma,N) &\simeq 1, \\ \Theta_j(E,K,\sigma,N) &\simeq 2\pi\ell, \qquad \ell\in\mathbb{Z}. \end{aligned} \tag{81}$$

This criterion refers to the channel-resolved quantity $g(E,K,\sigma,N)$, not yet to the transverse-integrated conductance.

C. From fixed-K resonances to integrated ridges

A ridge in $G^{(g)}(E,N)$ is a feature of the integrated quantity in Eq. (**??**). Therefore a fixed-K resonance survives transverse integration only if neighboring open channels resonate at nearly the same energy. Let $E_{\ell j}(K,N;\sigma)$ be the solution of

$$\Theta_j(E,K,\sigma,N) = 2\pi\ell \tag{82}$$

for a chosen branch. The open incident-channel window is

$$\mathcal{W}_{\rm in}(E,\sigma) = \{K\in[-\pi,\pi] : \chi_{\rm in}(E,K,\sigma)=1\}. \tag{83}$$

A practical K-flatness diagnostic is

$$\delta E_{\ell j}(N;\sigma) = \max_{K\in\mathcal{W}_{\rm in}} E_{\ell j}(K,N;\sigma) - \min_{K\in\mathcal{W}_{\rm in}} E_{\ell j}(K,N;\sigma). \tag{84}$$

If

$$\delta E_{\ell j}(N;\sigma) \lesssim \Gamma_{\ell j}(N;\sigma), \tag{85}$$

where $\Gamma_{\ell j}$ is the numerical linewidth of the corresponding conductance peak, then neighboring transverse channels reinforce each other and the branch is expected to appear as a visible ridge in $G(E,N)$. If $\delta E_{\ell j} \gg \Gamma_{\ell j}$, the fixed-$K$ resonances are dispersed over energy and are washed out by the transverse integral.

A residue-weighted diagnostic is obtained from

$$w_j(E,K,\sigma,N) = \frac{|C_j(E,K,\sigma,N)|^2}{|1-\mu_j(E,K,\sigma,N)|^2}. \tag{86}$$

This avoids overemphasizing formally resonant branches that are dark because their injection or hole-readout residue is negligible.

D. Numerical workflow for ridge validation

The formal interpretation becomes a testable result only after the predicted resonance branches are compared with the numerical conductance maps. The code should output, for each g, E, K, σ, and N:

(i) the open-channel mask $\chi_{\rm in}$;

(ii) R_N, R_A, $T_{\rm QP}$, and the unitarity error $1-R_N-R_A-T_{\rm QP}$;

(iii) the fixed-channel conductance $g(E,K,\sigma,N)$ and integrated conductance $G^{(g)}(E,N)$;

(iv) the round-trip matrix $\mathsf{M}_{\rm rt}^{(g)}$ and $s_{\min}(\mathbb{I}_4 - \mathsf{M}_{\rm rt}^{(g)})$;

(v) the eigenvalues $\mu_j = \rho_j e^{i\Theta_j}$, with phases unwrapped across the grid;

(vi) the residues C_j and weights w_j;

(vii) the branch curves $E_{\ell j}(K,N;\sigma)$ and the flatness measure $\delta E_{\ell j}(N;\sigma)$.

The central figure supporting the resonance interpretation should overlay the predicted curves $E_{\ell j}(N)$ on the computed $G_A(E,N)$ map, retaining only branches with $\rho_j \simeq 1$, appreciable $|C_j|$, and sufficient K-flatness. The round-trip eigenvector should then be used to diagnose whether a visible ridge is predominantly normal Fabry–Pérot-like, Andreev-like, or a mixed electron-hole interference mode.

VII. NUMERICAL RESULTS AND RESONANCE ANALYSIS

The formalism developed above expresses the differential conductance in terms of the microscopic Andreev-reflection amplitude. In the subgap regime, $|E| < \Delta_0$, no propagating quasiparticle mode exists in the superconducting lead, and the conductance reduces to

$$G_A^{(g)}(E,N) = \frac{1}{2\pi}\sum_{\sigma=\pm 1}\int_{-\pi}^{\pi} dK\,\chi_{\rm in}(E,K,\sigma)\,2\left|r_{he}^{(g)}(E,K,\sigma,N)\right|^2. \tag{87}$$

The purpose of this section is to evaluate Eq. (??) for the two contact geometries and to relate the resulting conductance ridges to the round-trip matrix

$$\mathcal{M}_{\rm rt}^{(g)} = R_F P_-(N) R_S^{(g)} P_+(N). \tag{88}$$

A. Numerical parameters and consistency checks

Unless otherwise stated, all energies are measured in units of the intralayer hopping t. The interlayer hopping is denoted by $t_\perp$, the superconducting gap by Δ_0, and the exchange field in the ferromagnetic lead by h. The central bilayer contains N longitudinal cells, and the transverse momentum $K \in [-\pi,\pi]$ is integrated by numerical quadrature.

For each point (E,K,σ,N), we compute the reflection and transmission probabilities

$$R_N = |r_{ee}|^2, \qquad R_A = |r_{he}|^2, \qquad T_{\rm QP} = \sum_{\nu\in{\rm prop.}} |t_\nu|^2. \tag{89}$$

The numerical solution is accepted only when the flux-conservation residual

$$\delta_{\rm flux} = |1 - R_N - R_A - T_{\rm QP}| \tag{90}$$

remains close to machine precision over the open-channel region $\chi_{\rm in} = 1$.

B. Andreev conductance maps

Figure ?? shows the subgap Andreev conductance $G_A^{(g)}(E,N)$ for the two contact geometries. The maps display a sequence of sharp ridge-like structures as a function of the excitation energy E and the bilayer length N. These ridges are the most prominent finite-size effect of the normal bilayer region.

The two geometries share the same central bilayer spectrum, but differ in the right-interface selector matrices and therefore in the superconducting reflection matrix $R_S^{(g)}$. Consequently, the ridge pattern is not determined by the bilayer dispersion alone. It also depends on how each central mode couples to the superconducting lead.

C. Fixed-K channel resonances

The integrated conductance $G_A(E,N)$ is obtained only after summing over spin and integrating over transverse momentum. The microscopic resonance condition, however, is naturally formulated at fixed (K,σ). We therefore first examine the channel-resolved conductance

$$g_A^{(g)}(E,K,\sigma,N) = \chi_{\rm in}(E,K,\sigma) 2\left|r_{he}^{(g)}(E,K,\sigma,N)\right|^2. \tag{91}$$

At fixed K, sharp maxima of Eq. (??) can be traced to the eigenvalues of the round-trip matrix $\mathcal{M}_{\rm rt}^{(g)}(E,K,\sigma,N)$. This fixed-channel analysis is the first step toward understanding which microscopic resonances survive in the integrated conductance.

D. Round-trip eigenvalues and bright resonances

Let

$$\mathcal{M}_{\rm rt}^{(g)} v_j^R = \mu_j v_j^R, \qquad (v_j^L)^\dagger \mathcal{M}_{\rm rt}^{(g)} = \mu_j (v_j^L)^\dagger, \tag{92}$$

with biorthogonal normalization

$$(v_j^L)^\dagger v_k^R = \delta_{jk}. \tag{93}$$

For a diagonalizable round-trip matrix, the Andreev amplitude can be written as

$$r_{he}^{(g)} = \sum_j \frac{C_j^{(g)}}{1-\mu_j^{(g)}}, \tag{94}$$

where

$$\begin{aligned} C_j^{(g)} &= \left[e_h^\dagger T_{F\leftarrow C} P_- R_S^{(g)} P_+ v_j^R\right] \\ &\quad\times \left[(v_j^L)^\dagger t_{\rm in}\right]. \end{aligned} \tag{95}$$

Writing

$$\mu_j^{(g)} = \rho_j^{(g)} e^{i\Theta_j^{(g)}}, \tag{96}$$

a single isolated resonance occurs when

$$\rho_j^{(g)} \simeq 1, \qquad \Theta_j^{(g)} \simeq 2\pi\ell, \qquad \ell\in\mathbb{Z}. \tag{97}$$

The modulus ρ_j measures how much of the mode survives one complete round trip, while Θ_j is the accumulated round-trip phase. The residue C_j determines whether the pole is bright or dark in the physical conductance: a near-zero denominator in Eq. (??) produces a visible ridge only when the incident electron couples appreciably to the corresponding round-trip eigenmode and that eigenmode has appreciable hole readout at the ferromagnetic lead.

A useful real-axis diagnostic of resonance proximity is the smallest singular value

$$s_{\rm min}^{(g)} = s_{\rm min}\left[I_4 - \mathcal{M}_{\rm rt}^{(g)}\right]. \tag{98}$$

Unlike the determinant, this quantity measures near-singularity directly and is therefore more robust when several round-trip eigenvalues contribute simultaneously.

E. From fixed-K resonances to integrated ridges

The condition in Eq. (??) is a fixed-K criterion. A ridge in the integrated conductance appears only if the corresponding resonance branch remains sufficiently aligned as K varies over the open-channel window. Let

$$E_{\ell j}^{(g)}(K, N; \sigma) \tag{99}$$

be a solution of

$$\Theta_j^{(g)}(E, K, \sigma, N) = 2\pi\ell. \tag{100}$$

A simple measure of transverse dispersion is

$$\begin{aligned} \delta E_{\ell j}^{(g)}(N; \sigma) = \max_{K \in \mathcal{W}_{\text{in}}} E_{\ell j}^{(g)}(K, N; \sigma) \\ - \min_{K \in \mathcal{W}_{\text{in}}} E_{\ell j}^{(g)}(K, N; \sigma), \end{aligned} \tag{101}$$

where

$$\mathcal{W}_{\text{in}}(E, \sigma) = \{K \in [-\pi, \pi] : \chi_{\text{in}}(E, K, \sigma) = 1\}. \tag{102}$$

Branches with

$$\delta E_{\ell j}^{(g)}(N; \sigma) \lesssim \Gamma_{\ell j}^{(g)}(N) \tag{103}$$

are expected to survive transverse integration, where $\Gamma_{\ell j}^{(g)}(N)$ is the numerical linewidth of the associated fixed-K resonance. Branches that disperse strongly with K are washed out by the transverse integral even if they are sharp in a single channel.

This comparison is the central test of the resonance interpretation. A visible ridge must satisfy three conditions simultaneously: the round-trip matrix must be nearly singular, the relevant modal residue must be large enough to make the pole bright, and the fixed-K resonance branch must be flat enough to survive the transverse integral.

F. Comparison between the two geometries

The two geometries differ only through the superconducting-side interface maps, but this local change is sufficient to modify the full round-trip matrix and therefore the resonance pattern. In the $1 \to 1$ geometry, the ferromagnetic and superconducting monolayers are connected through the same graphene layer. In the $1 \to 2$ geometry, a quasiparticle injected from the ferromagnet must couple through the bilayer region before reaching the layer connected to the superconductor. The resulting change in $R_S^{(g)}$ and in the modal residues $C_j^{(g)}$ changes which resonances are bright in the integrated conductance.

G. Summary of the numerical results

The conductance maps reveal that the finite bilayer does not merely act as a featureless spacer between the ferromagnet and the superconductor. Instead, it behaves as a coherent cavity whose internal modes undergo repeated reflection between the two atomically distinct contacts. The round-trip matrix $\mathcal{M}_{\text{rt}}^{(g)}$ provides the microscopic description of this cavity. Its eigenvalues determine the resonance phase and lifetime, while the residues determine whether the corresponding poles are visible in the Andreev conductance. The ridge overlays therefore provide a direct connection between the transfer-matrix formalism and the structures observed in $G_A(E, N)$.

VIII. CONCLUSIONS

This section will be finalized after the numerical maps and ridge overlays are inserted. The intended conclusion is that an atomically resolved transfer-matrix treatment of a ferromagnet–bilayer-graphene–superconductor junction produces a compact round-trip description of Andreev transport. The finite bilayer acts as a coherent cavity whose resonant structure is controlled by the eigenvalues and residues of $\mathsf{M}_{\text{rt}}^{(g)}$, while the observed conductance ridges additionally require sufficient transverse-channel coherence to survive the K integration.

The final version should emphasize three points: first, the interface transparency is not fitted but follows from the lattice connectivity and layer termination; second, the stable propagation basis avoids direct ill-conditioned powers of the central transfer matrix; third, the $1 \to 1$ and $1 \to 2$ geometries provide distinct microscopic routes for Andreev reflection because the superconducting contact samples different layer components of the bilayer modes.

Appendix A: BdG commutators and pairing convention

This appendix records the convention needed to keep the two spin-singlet BdG sectors in the same algebraic form. In a complete one-particle basis, including spin, the pairing matrix is antisymmetric,

$$\Delta_{ij} = -\Delta_{ji}. \tag{A1}$$

For onsite s-wave pairing the spatial, layer, and sublattice labels coincide, while the two spin labels are opposite; the spatial part is symmetric, but the full spin-orbital matrix remains antisymmetric.

For the general quadratic pairing term

$$\hat{H}_\Delta = \frac{1}{2} \sum_{ij} \left(\Delta_{ij} \hat{c}_i^\dagger \hat{c}_j^\dagger + \Delta_{ij}^* \hat{c}_j \hat{c}_i \right), \tag{A2}$$

the creation part gives

$$\left[\frac{1}{2} \sum_{ij} \Delta_{ij} \hat{c}_i^\dagger \hat{c}_j^\dagger, v_l \hat{c}_l \right] = \frac{1}{2} \sum_i v_l \left(\Delta_{il} - \Delta_{li} \right) \hat{c}_i^\dagger \tag{A3}$$

$$= \sum_i \Delta_{il} v_l \hat{c}_i^\dagger, \tag{A4}$$

where Eq. (**??**) was used. The conjugate pairing part gives a contribution proportional to

$$\sum_i \Delta_{il}^* u_i \hat{c}_l. \tag{A5}$$

Thus the index order in the hole equation is fixed by

$$(\Delta^\dagger)_{li} = \Delta_{il}^*. \tag{A6}$$

With the quasiparticle operator of Eq. (**??**), the s_σ factor in the hole component cancels the spin-dependent sign of the singlet commutator. After this cancellation the sector equations can be written as

$$Eu_k = \sum_l h_{kl} u_l + \sum_i \Delta_{ki} v_i, \tag{A7}$$

$$Ev_l = -\sum_i h_{il} v_i + \sum_i \Delta_{il}^* u_i. \tag{A8}$$

Equation (**??**) is the component form of the BdG block matrix in Eq. (**??**).

Appendix B: Transfer matrices and bilayer roots

This appendix stores the algebra that is too detailed for the main text. For a normal monolayer with scalar energy parameter x, the staggered state obeys

$$L_M \phi_{m+1} = R_M \phi_m, \tag{B1}$$

with

$$L_M = \begin{pmatrix} x & \xi \\ \eta^* & 0 \end{pmatrix}, \qquad R_M = \begin{pmatrix} 0 & -\eta \\ -\xi^* & -x \end{pmatrix},$$
$$T_M = L_M^{-1} R_M. \tag{B2}$$

For the normal AB bilayer with $x_j = \epsilon - V_j$, the recurrence is

$$L_{\mathrm{BL}} \Psi_{m+1} = R_{\mathrm{BL}} \Psi_m, \qquad T_{\mathrm{BL}} = L_{\mathrm{BL}}^{-1} R_{\mathrm{BL}}, \tag{B3}$$

where

$$L_{\mathrm{BL}} = \begin{pmatrix} x_1 & \xi & 0 & 0 \\ \eta^* & 0 & 0 & 0 \\ 0 & 0 & x_2 & \xi \\ t_\perp & 0 & \eta^* & 0 \end{pmatrix}, \tag{B4}$$

$$R_{\mathrm{BL}} = \begin{pmatrix} 0 & -\eta & 0 & -t_\perp \\ -\xi^* & -x_1 & 0 & 0 \\ 0 & 0 & 0 & -\eta \\ 0 & 0 & -\xi^* & -x_2 \end{pmatrix}. \tag{B5}$$

A useful implementation check is

$$\det T_{\mathrm{BL}} = \left(\frac{\eta}{\eta^*}\right)^2 = e^{2iK}, \tag{B6}$$

valid away from $\eta(K) = 0$.

Using the Bloch form $A_j(m) = U_{Aj} e^{im\vartheta}$, $B_j(m) = U_{Bj} e^{im\vartheta}$ and defining

$$f(\vartheta, K) = \eta(K) + \xi e^{i\vartheta}, \qquad s = |f|^2, \tag{B7}$$

the bilayer secular equation is

$$(x_1^2 - s)(x_2^2 - s) - t_\perp^2 x_1 x_2 = 0. \tag{B8}$$

The two solutions are

$$s_\nu(\epsilon) = \frac{1}{2}\left[x_1^2 + x_2^2 + \nu\sqrt{(x_1^2 - x_2^2)^2 + 4t_\perp^2 x_1 x_2}\right], \quad \nu = \pm 1. \tag{B9}$$

Writing $\vartheta = K/2 + Q$, one obtains

$$s = t^2 \left[3 + 2\cos K + 4\cos(K/2)\cos Q\right], \tag{B10}$$

which leads to the roots quoted in Eq. (**??**).

Appendix C: Lattice current and pseudo-unitarity

For a hopping bond crossing the cut between cells m and $m+1$, the particle current carried by an electron amplitude is

$$J_{m\to m+1} = 2\,\mathrm{Im}\left[\xi A^*(m) B(m+1)\right]. \tag{C1}$$

This sign follows from the continuity equation with the hopping convention of Eq. (**??**). For a BdG vector, the hole contribution enters with the opposite sign because a hole amplitude represents a missing electron. This gives the flux metrics in Eqs. (**??**) and (**??**).

If the current is conserved by one transfer step, then

$$(T\Phi)^\dagger \Sigma (T\Phi) = \Phi^\dagger \Sigma \Phi \tag{C2}$$

for all states Φ, and hence

$$T^\dagger \Sigma T = \Sigma. \tag{C3}$$

For transfer eigenvectors this implies

$$(\lambda_i^* \lambda_j - 1)\, \zeta_i^\dagger \Sigma \zeta_j = 0. \tag{C4}$$

In particular, an evanescent mode with $|\lambda| \neq 1$ has zero self-flux, while distinct propagating modes can be chosen flux-orthogonal. Inside a degenerate propagating subspace, a stable numerical procedure is to diagonalize the Hermitian matrix $Z^\dagger \Sigma Z$ formed from any computed basis Z for the subspace, then normalize the resulting combinations to unit flux.

Appendix D: Interface matching systems

The local interface maps used in Sec. **??** are obtained from selector matrices that extract the active layer and the terminated-layer components from the central state Ξ. At the left contact, layer 1 continues into the ferromagnetic lead and layer 2 terminates. Let S_L extract the four layer-1 BdG components and M_L extract the two independent missing-neighbor components of the terminated layer. Then the homogeneous left-interface problem is

$$\begin{pmatrix} S_L V_+ & -Z_F^- \\ M_L V_+ & 0_{2\times 2} \end{pmatrix} \begin{pmatrix} a_+^L \\ r \end{pmatrix} = - \begin{pmatrix} S_L V_- \\ M_L V_- \end{pmatrix} a_-^L. \tag{D1}$$

Adding the physical incident state replaces the right-hand side of the active-layer equation by $\zeta_{\text{in}} - S_L V_- a_-^L$, leading to the affine map in Eq. (**??**).

At the right contact, the active selector depends on the geometry. For $g = 1 \to 1$, the superconducting lead continues from layer 1; for $g = 1 \to 2$, it continues from layer 2. The right-interface system is

$$\begin{pmatrix} S_R^{(g)} V_- & -Z_S^+ \\ M_R^{(g)} V_- & 0_{2\times 2} \end{pmatrix} \begin{pmatrix} a_-^R \\ t \end{pmatrix} = - \begin{pmatrix} S_R^{(g)} V_+ \\ M_R^{(g)} V_+ \end{pmatrix} a_+^R. \tag{D2}$$

The full 6×6 systems in Eqs. (**??**) and (**??**), solved with pivoted linear algebra, are the numerical definition of the contact maps. Schur-complement reductions are useful analytically, but a singular chosen pivot does not by itself imply that the full contact problem is singular.

[1] H. Z. Olyaei, P. Ribeiro, and E. V. Castro, "Transmission across a bilayer graphene region," Phys. Rev. B **99**, 205436 (2019).

[2] G. E. Blonder, M. Tinkham, and M. T. Klapwijk, "Transition from metallic to tunneling regimes in superconducting microconstrictions," Phys. Rev. B **25**, 4515 (1982).

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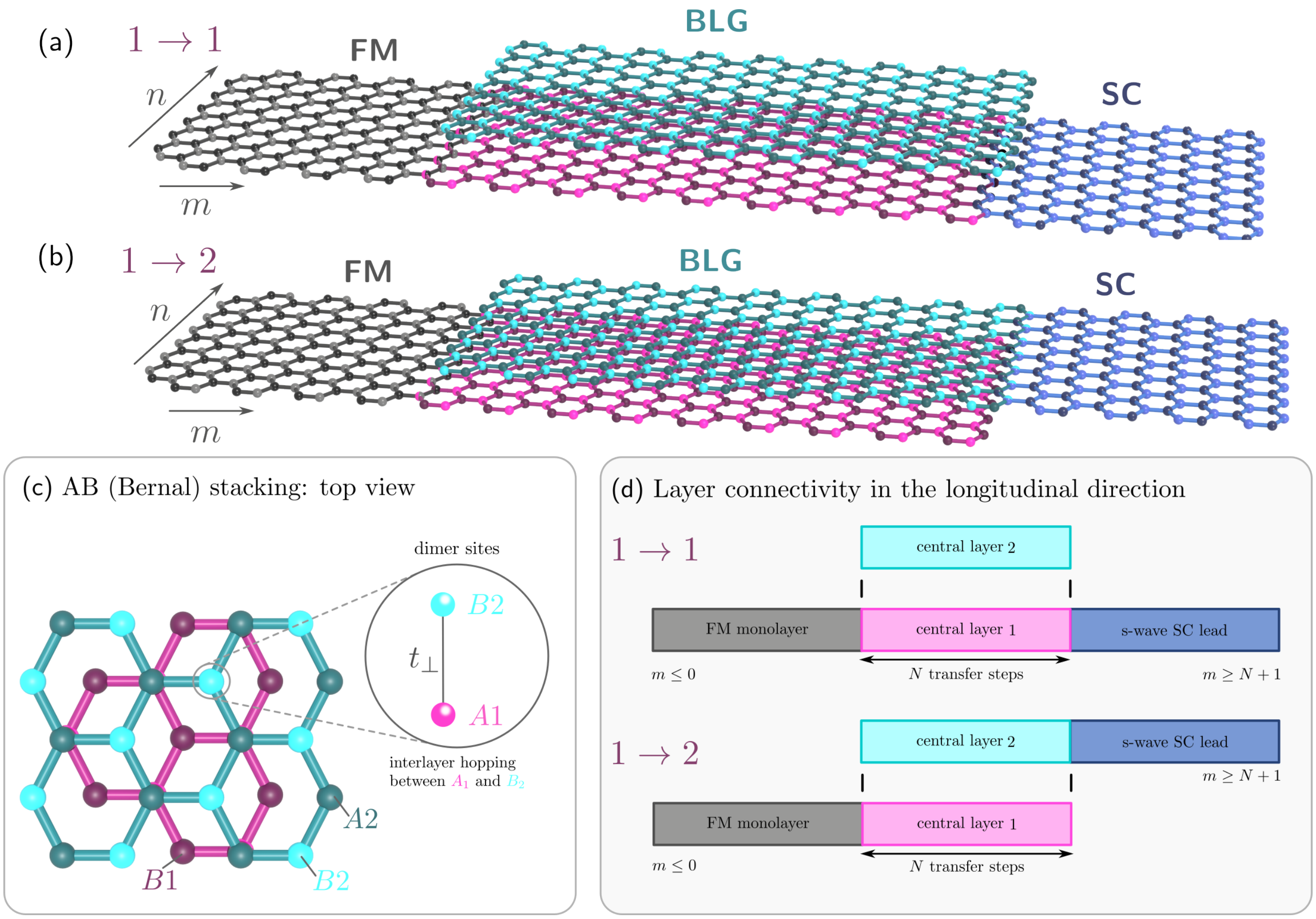

Figure 1. Geometry, lattice structure, and longitudinal connectivity of the ferromagnet–bilayer-graphene–superconductor (FM–BLG–SC) junction. (a) Atomic representation of the $1 \to 1$ geometry, in which the ferromagnetic (FM) and superconducting (SC) monolayer graphene leads are connected to the same layer of the finite bilayer graphene (BLG) region. (b) Corresponding $1 \to 2$ geometry, in which the FM and SC leads are connected to opposite graphene layers. The longitudinal and transverse lattice directions are denoted by m and n, respectively. Gray, magenta, cyan, and indigo identify the FM monolayer, central layer 1, central layer 2, and SC monolayer, respectively. (c) Top view of the AB (Bernal) stacking in the central bilayer. Different shades within each layer distinguish the A and B sublattices. The enlarged detail shows the vertically registered A_1 and B_2 dimer sites, coupled by the interlayer hopping $t_\perp$; the A_1 site is hidden beneath B_2 in the top projection. The non-dimer sites B_1 and A_2 are also indicated. (d) Longitudinal layer-connectivity representation of the $1 \to 1$ and $1 \to 2$ junctions. The FM lead occupies $m \leq 0$, the finite BLG region extends over N transfer steps, $1 \leq m \leq N$, and the SC lead occupies $m \geq N+1$. In the $1 \to 1$ geometry the SC lead continues from central layer 1, whereas in the $1 \to 2$ geometry it continues from central layer 2.

figures/results/andreev_maps.pdf

Figure 2. Subgap Andreev conductance maps for the two contact geometries. Panel (a) shows $G_A^{(1\to 1)}(E, N)$, while panel (b) shows $G_A^{(1\to 2)}(E, N)$. The color scale represents the dimensionless conductance per transverse unit cell, in units of e^2/h. The ridge-like structures originate from coherent round trips inside the finite bilayer region.

figures/results/fixed_k_resonances.pdf

Figure 3. Representative fixed-K Andreev conductance resonances. Unlike the integrated map, this quantity probes a single transverse channel and therefore exposes the microscopic resonances before transverse averaging.

figures/results/round_trip_diagnostics.pdf

Figure 4. Round-trip resonance diagnostics for a representative geometry and spin sector. The plotted quantities may include $s_{\min}(I_4 - \mathcal{M}_{\mathrm{rt}})$, ρ_j, Θ_j, and the residue weight $|C_j|/|1-\mu_j|$. Visible conductance ridges are associated with round-trip eigenvalues close to unity and with non-small residues.

figures/results/ridge_overlays.pdf

Figure 5. Conductance maps with resonance-branch overlays. The background shows $G_A^{(g)}(E, N)$, while the curves show the predicted branches $E_{\ell j}^{(g)}(N)$ obtained from the round-trip phase condition. Agreement between the curves and the bright ridges identifies the ridges as coherent finite-length resonances of the bilayer cavity.